

Biodiversity and Conservation

An NCERT-Centric Deep Dive for NEET-UG

Core Concepts & Definitions

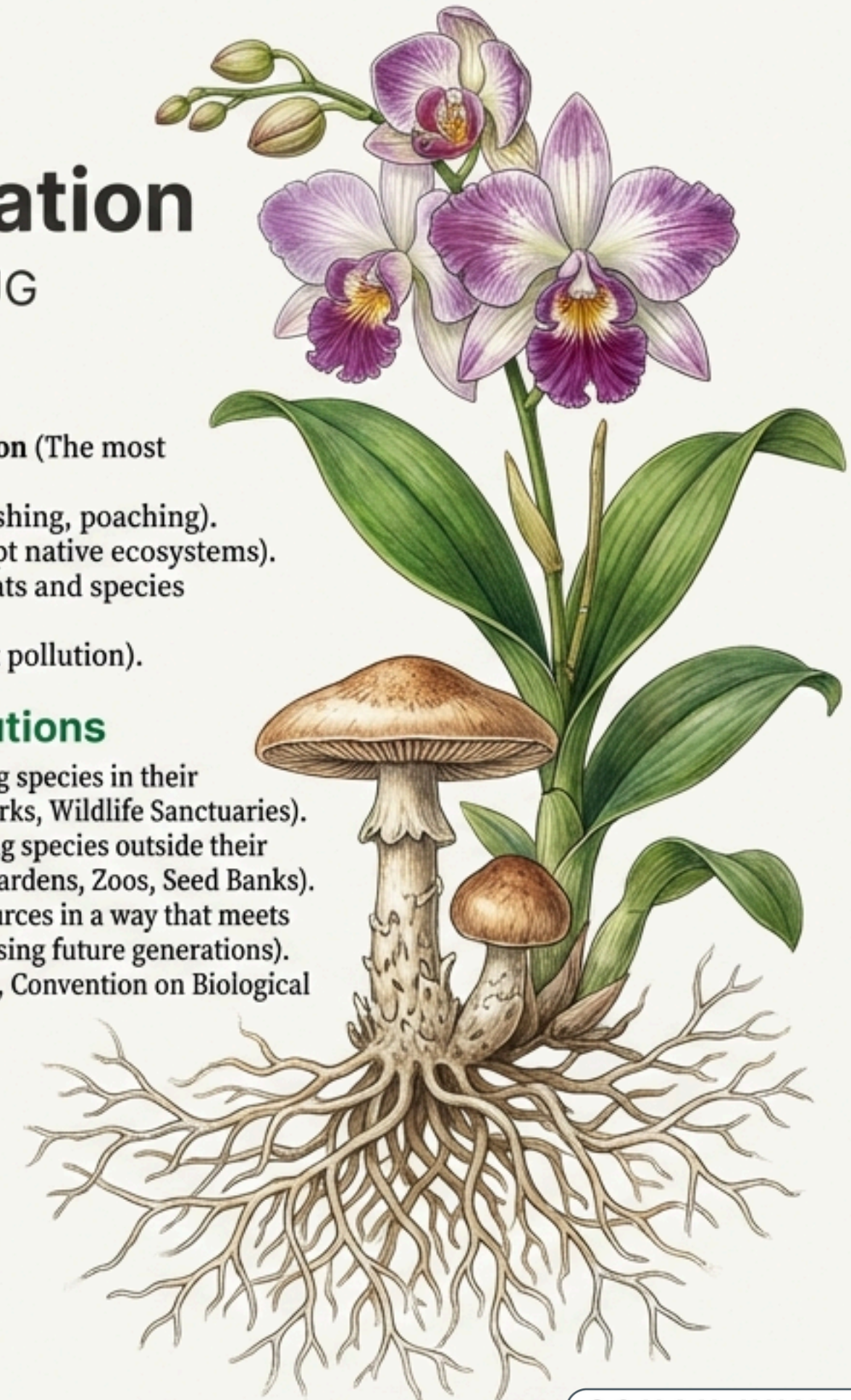
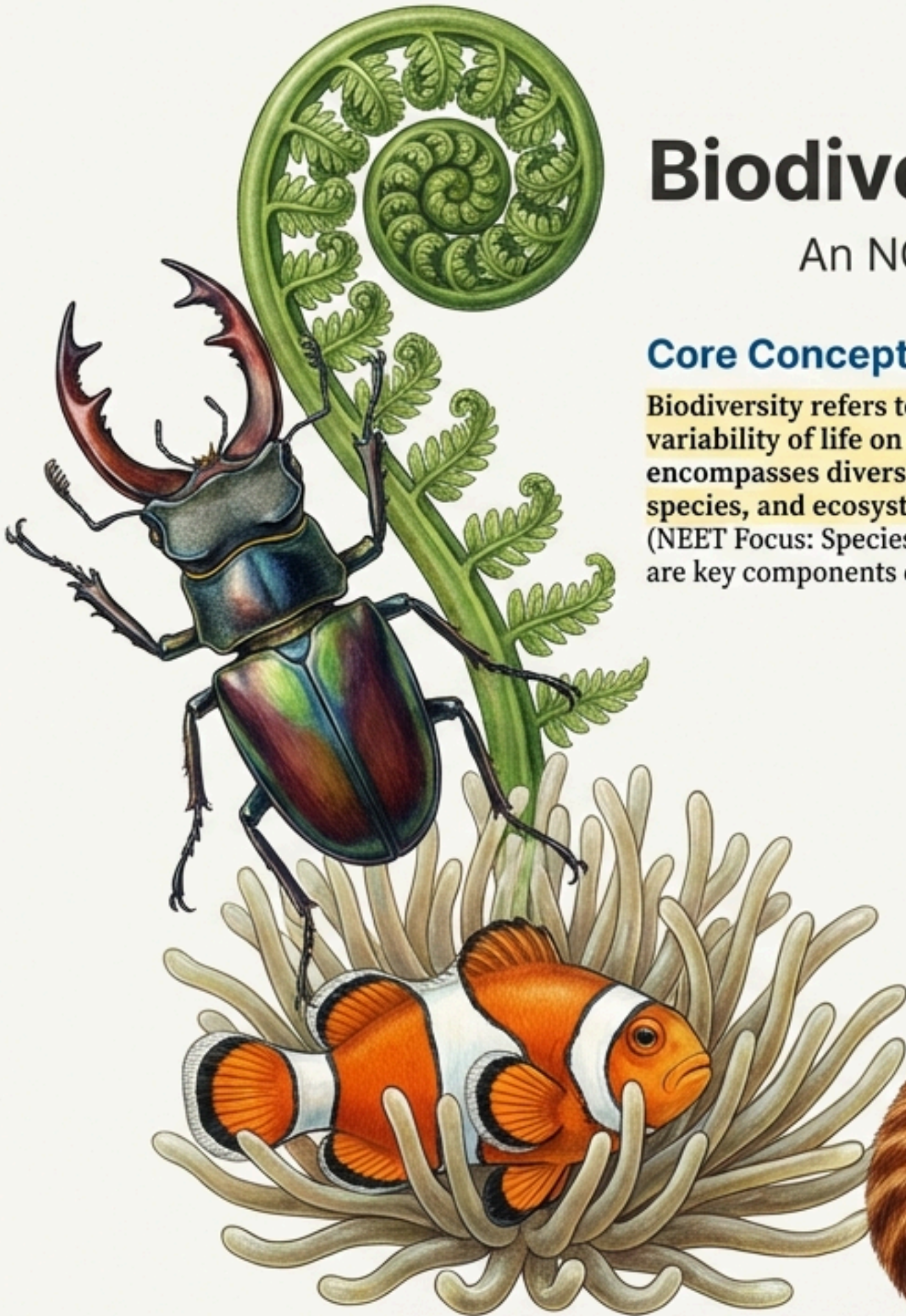
Biodiversity refers to the variety and variability of life on Earth. It encompasses diversity at the genetic, species, and ecosystem levels. (NEET Focus: Species richness and evenness are key components of diversity).

Threats & Losses

- Habitat loss and fragmentation (The most significant threat).
- Overexploitation (e.g., overfishing, poaching).
- Invasive alien species (Disrupt native ecosystems).
- Climate change (Alters habitats and species distributions).
- Pollution (Chemical and light pollution).

Conservation & Solutions

- In-situ conservation (Protecting species in their natural habitats, e.g., National Parks, Wildlife Sanctuaries).
- Ex-situ conservation (Protecting species outside their natural habitats, e.g., Botanical Gardens, Zoos, Seed Banks).
- Sustainable use (Utilizing resources in a way that meets current needs without compromising future generations).
- International agreements (e.g., Convention on Biological Diversity).



Chapter Overview: Biodiversity and Conservation

NCERT Syllabus Mapping

13.1 Biodiversity

13.1.1 How Many Species are there on Earth and How Many in India?

13.1.2 Patterns of Biodiversity

13.1.3 The importance of Species Diversity to the Ecosystem

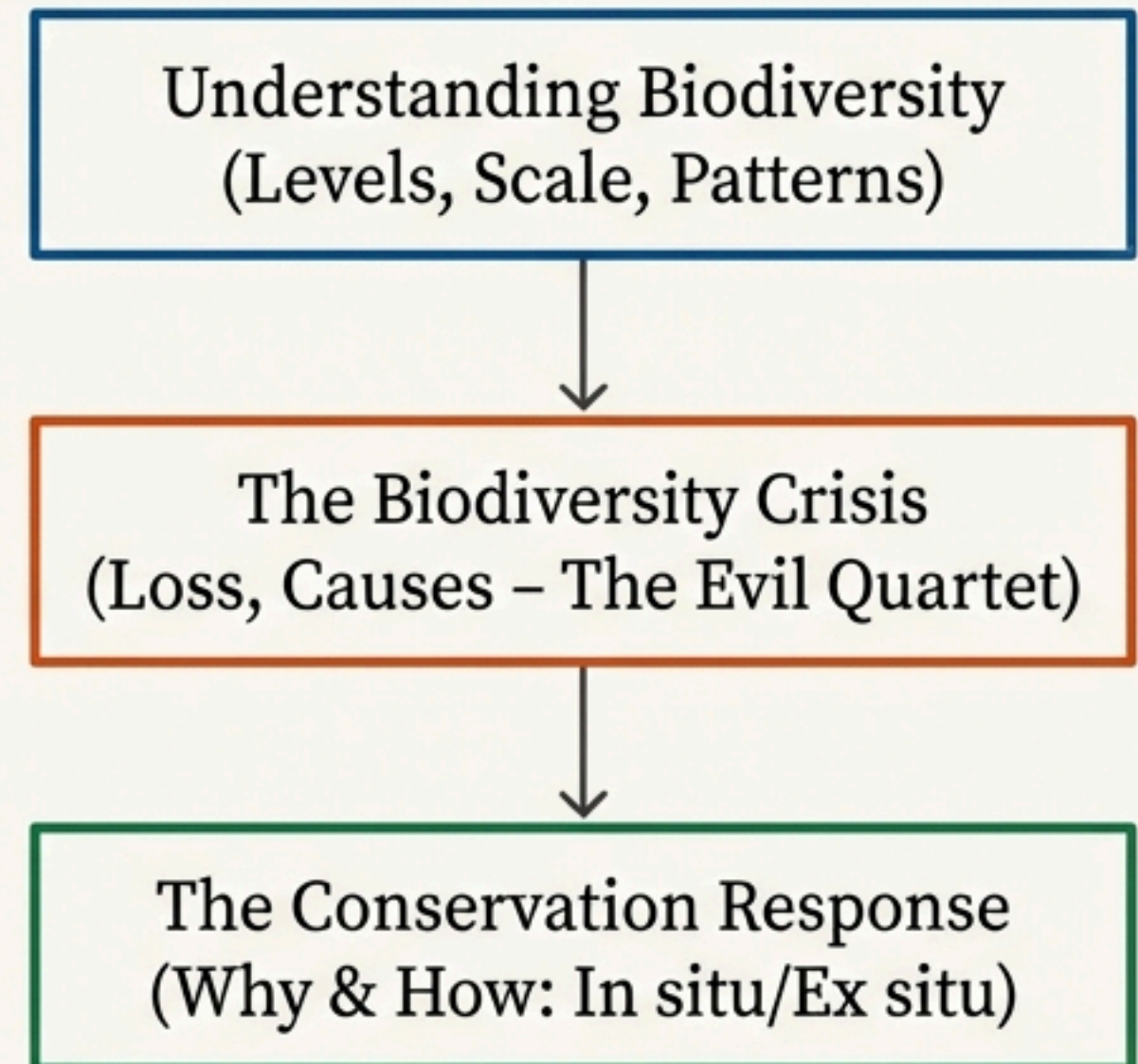
13.1.4 Loss of Biodiversity

13.2 Biodiversity Conservation

13.2.1 Why Should We Conserve Biodiversity?

13.2.2 How do we conserve Biodiversity?

Conceptual Flow



NEET Weightage Trend: High

13.1 Biodiversity: The Three Levels of Diversity

“Biodiversity is the term popularised by the sociobiologist **Edward Wilson** to describe the combined diversity at all the levels of biological organisation.”

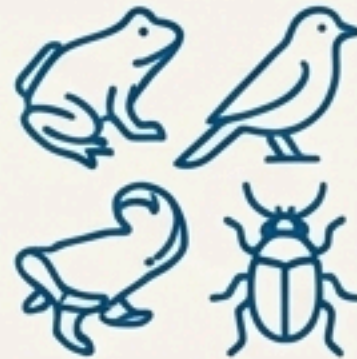


1. Genetic Diversity: Variation within a single species.

A single species might show high diversity at the genetic level over its distributional range.

The medicinal plant **Rauwolfia vomitoria** in the Himalayas shows genetic variation in the potency and concentration of the active chemical, **reserpine**.

India has over **50,000** genetically different strains of rice and **1,000** varieties of mango.



2. Species Diversity: Variety of species in a region.

The diversity at the species level.

The Western Ghats have a greater amphibian species diversity than the Eastern Ghats.



3. Ecological Diversity: Variety of ecosystems.

Diversity at the ecosystem level.

India, with its **deserts, rain forests, mangroves, and wetlands**, has a greater ecosystem **diversity** than a Scandinavian country like **Norway**.

13.1.1 The Scale of Life: How Many Species on Earth?

Global Estimates:

- Recorded Species (IUCN 2004): Slightly more than 1.5 million plant and animal species described.
- Scientifically Sound Estimate (Robert May): Global species diversity is placed at about 7 million.

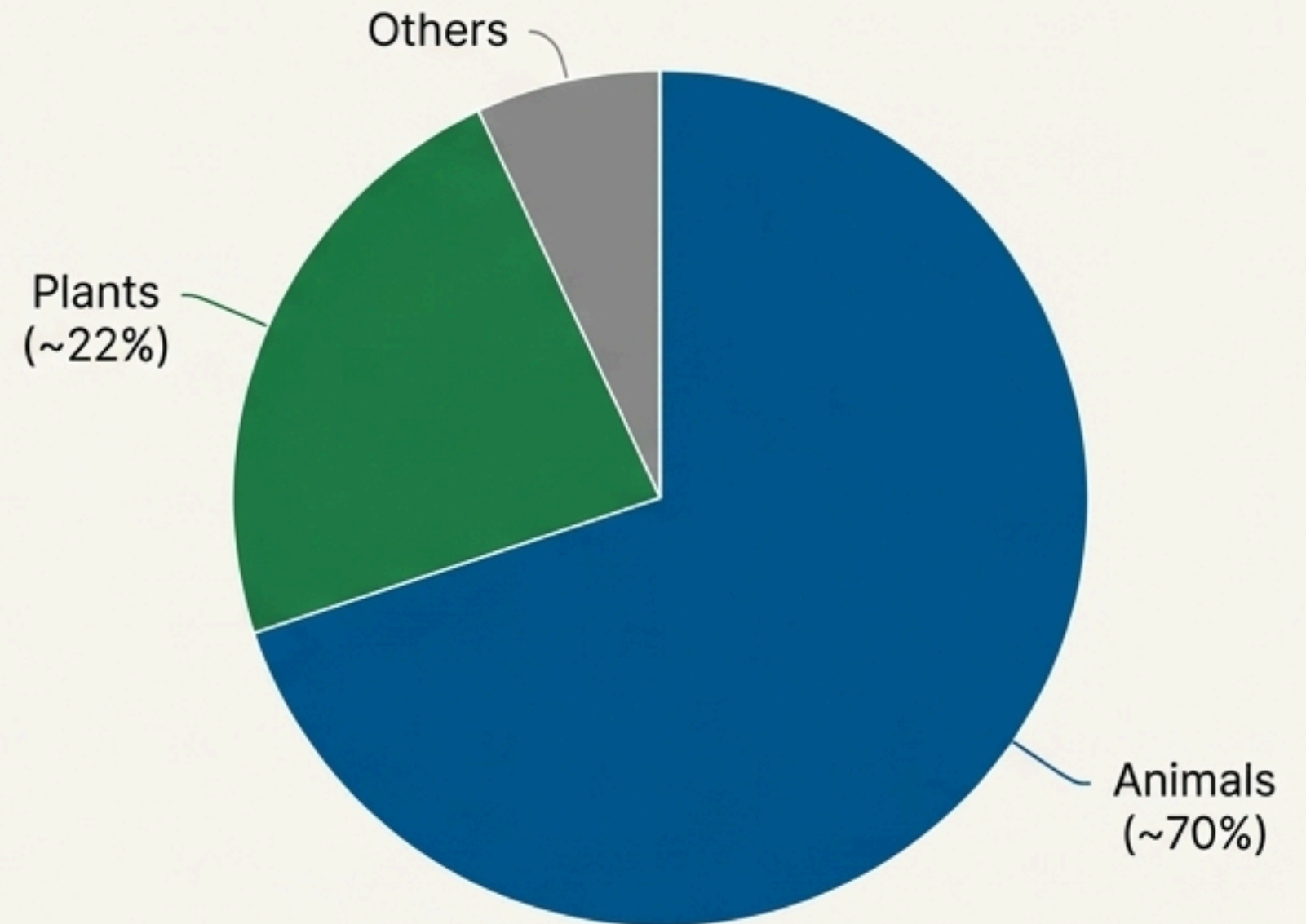
Taxonomic Distribution: The 70% Rule:

- More than 70% of all recorded species are animals.
- Plants (including algae, fungi, etc.) comprise no more than 22%.
- Among animals, insects are the most species-rich group, making up more than 70% of the total. (i.e., out of every 10 animals, 7 are insects).
- The number of fungi species is more than the combined total of the species of fishes, amphibians, reptiles, and mammals.

The Prokaryote Caveat:

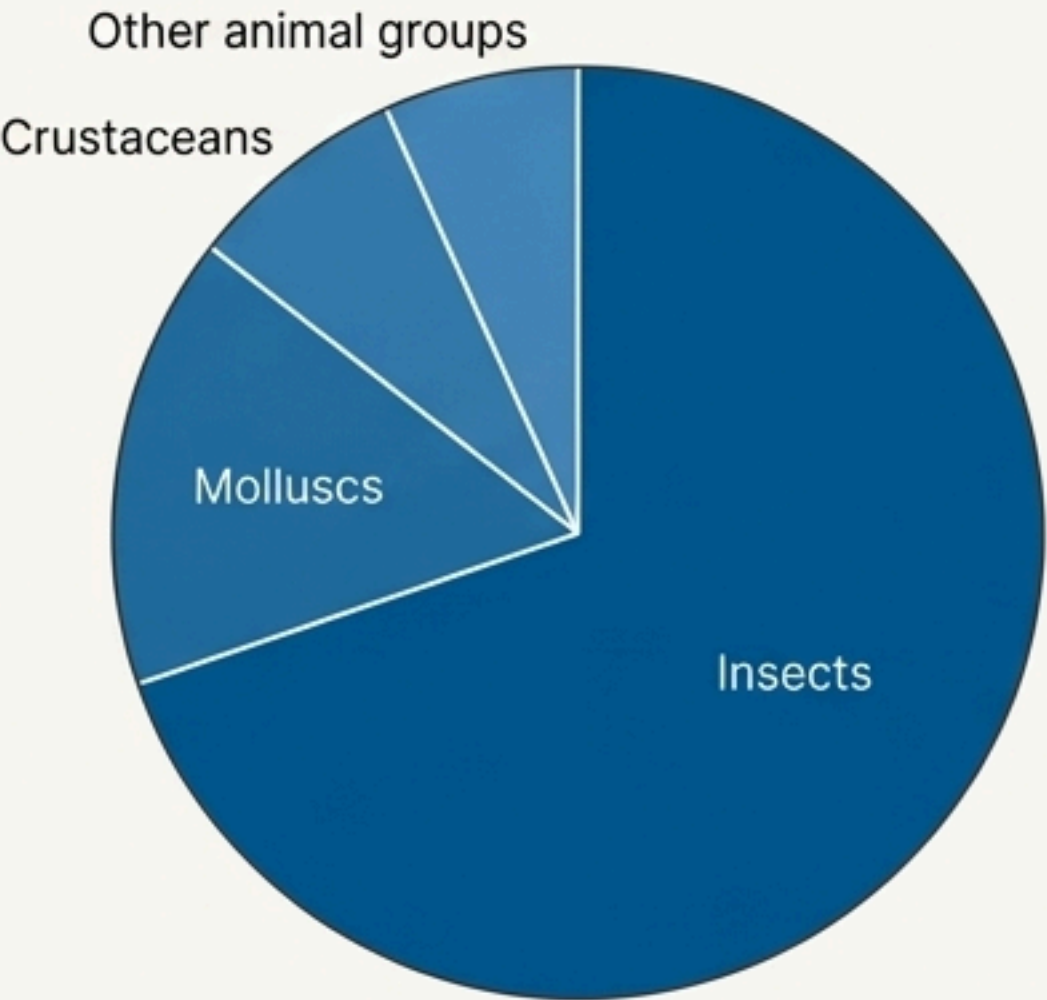
- These estimates do not include prokaryotes, as conventional taxonomy is unsuitable and many species are not culturable. Their true diversity could be in the millions.

Global Species Distribution

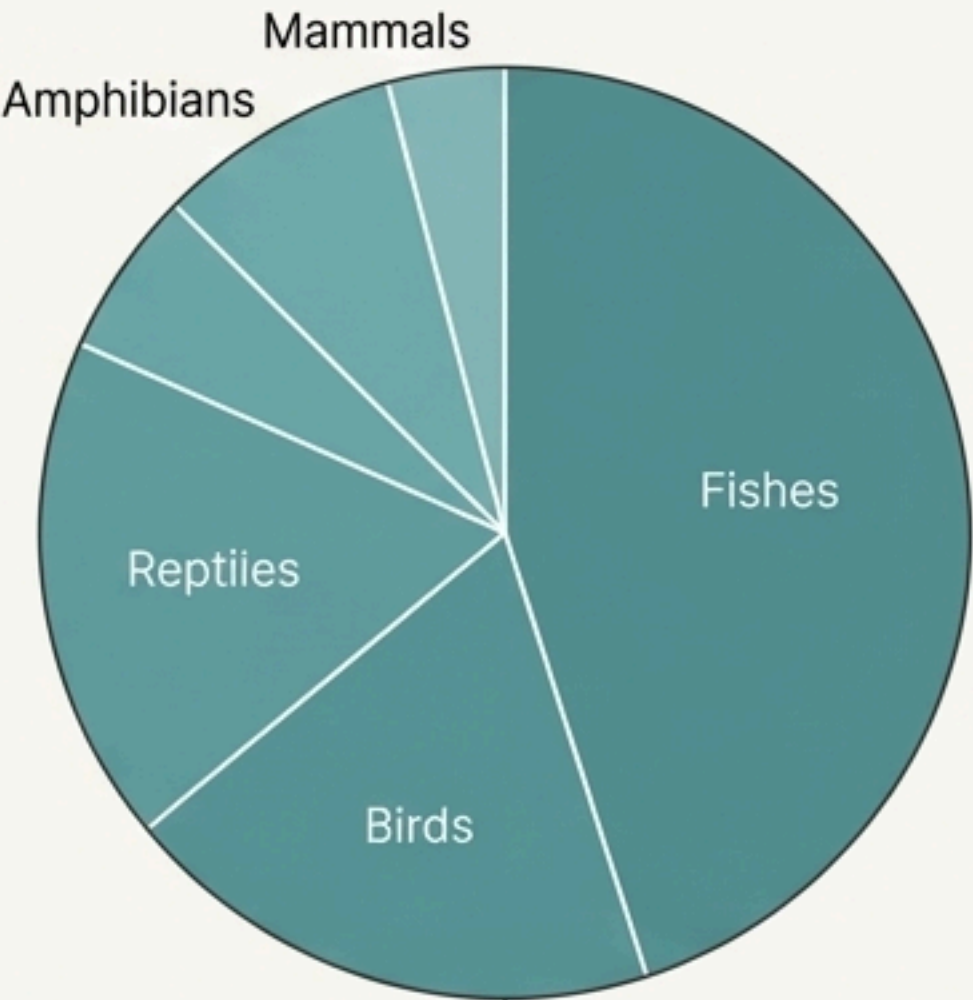


Visualizing Global Biodiversity: Proportionate Number of Species

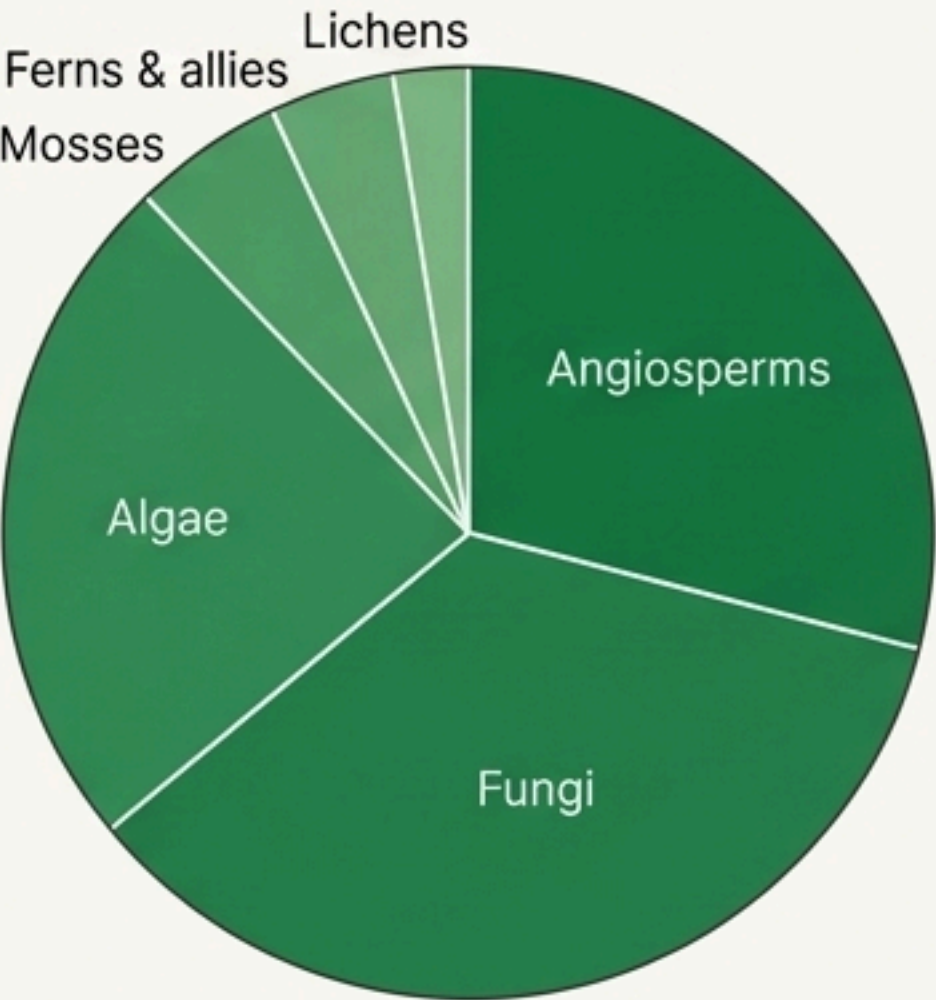
Invertebrates



Vertebrates



Plants



Examiner's Target

Relative abundance is key. Note that **Insects** dominate invertebrates, **Fishes** dominate vertebrates, and the number of **Fungi** species is greater than all vertebrate species combined.

India's Share in Global Species Diversity

2.4%

World's Land Area

8.1%

Global Species Diversity

This makes India one of the **12 mega diversity countries** of the world.

Recorded Species in India

- **Plants:** Nearly 45,000 species.
- **Animals:** Twice as many (approximately 90,000 species).

Projected Undiscovered Species (Applying May's Estimate)

If only 22% of total species have been recorded globally, India likely has:

- More than **1,00,000 plant species** yet to be discovered.
- More than **3,00,000 animal species** yet to be discovered.

“Nature’s biological library is burning even before we catalogued the titles of all the books stocked there.” - NCERT Quote in Source Serif Pro Italic

13.1.2 Patterns of Biodiversity (i) Latitudinal Gradients

Species diversity **decreases** as we move away from the **equator towards the poles**. Tropics (23.5° N to 23.5° S) harbour more species than temperate or polar areas.

Supporting Data

Region	Latitude	Bird Species
Colombia	Near Equator	~1,400
India	Tropical	>1,200
New York	41° N	105
Greenland	71° N	56

Case Study: Amazonian Rain Forest

The greatest biodiversity on Earth, home to >40,000 plant species, 3,000 fish species, and >1,25,000 invertebrate species.

Hypotheses for High Tropical Diversity

1. **Speciation as a function of time:** Tropical latitudes have remained undisturbed for millions of years (no frequent glaciations), allowing longer evolutionary time.
2. **Constant environments:** Tropical environments are less seasonal and more predictable, promoting niche specialisation.
3. **Higher Solar Energy:** More solar energy contributes to higher productivity, which in turn might contribute to greater diversity.

13.1.2 Patterns of Biodiversity (ii) Species-Area Relationships

Pioneer: Alexander von Humboldt, German naturalist and geographer.

Observation

Within a region, species richness increases with increasing explored area, but only up to a limit.

Relationship

Graphical Representation: A rectangular hyperbola on a normal scale.

Equation (on a logarithmic scale):

$$\log S = \log C + Z \log A$$

S = Species richness

A = Area

Z = Slope of the line (regression coefficient)

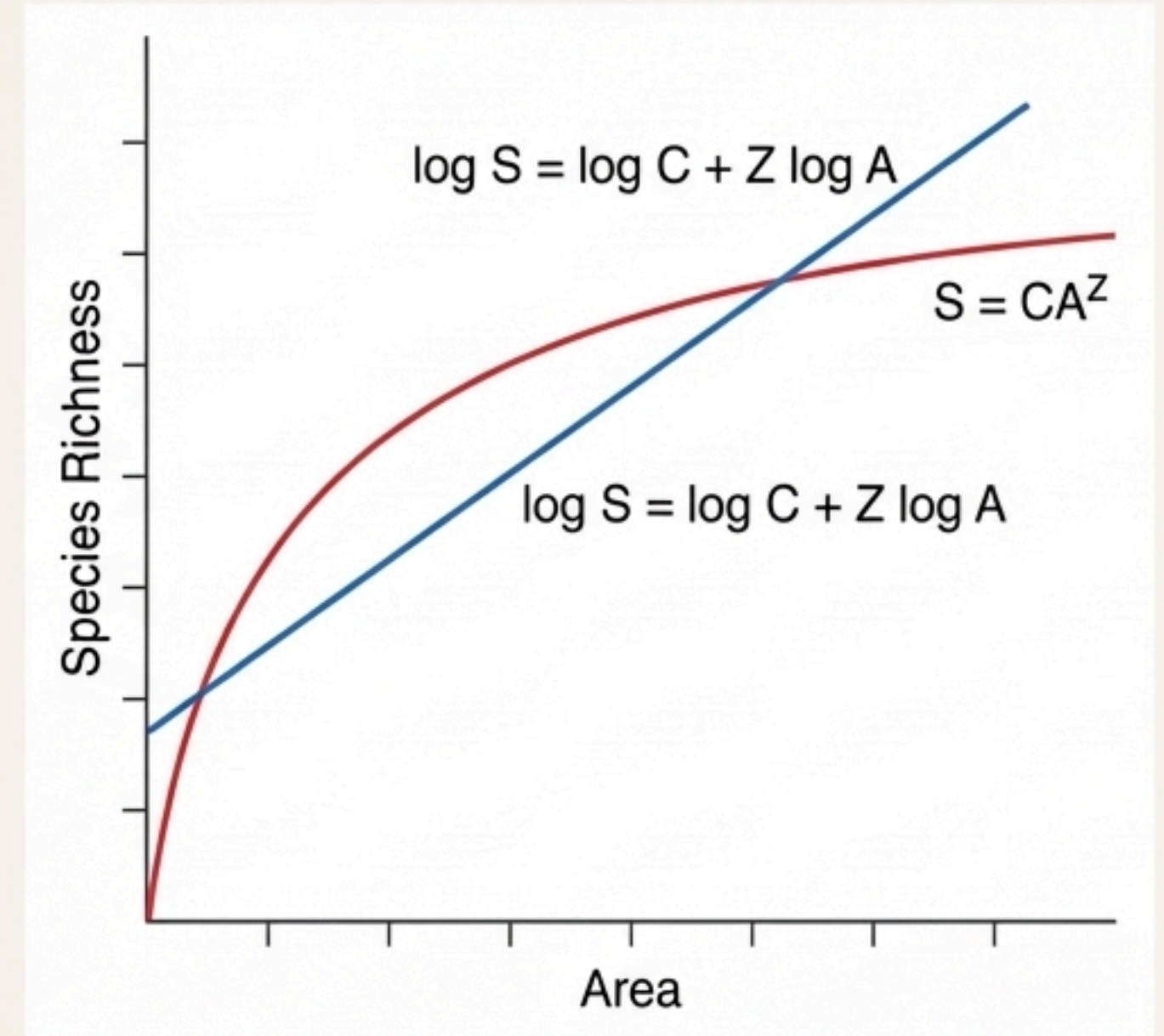
C = Y-intercept

Interpreting 'Z' (The Slope)

For regular regions (e.g., plants in Britain, birds in California), Z lies in the range of 0.1 to 0.2.

For very large areas (e.g., entire continents), the slope is much steeper, with Z values in the range of 0.6 to 1.2.

(Example: Frugivorous birds/mammals in tropical forests, $Z = 1.15$).



13.1.3 Why Species Richness Matters: Stability and the Rivet Popper Hypothesis

Ecosystem Stability

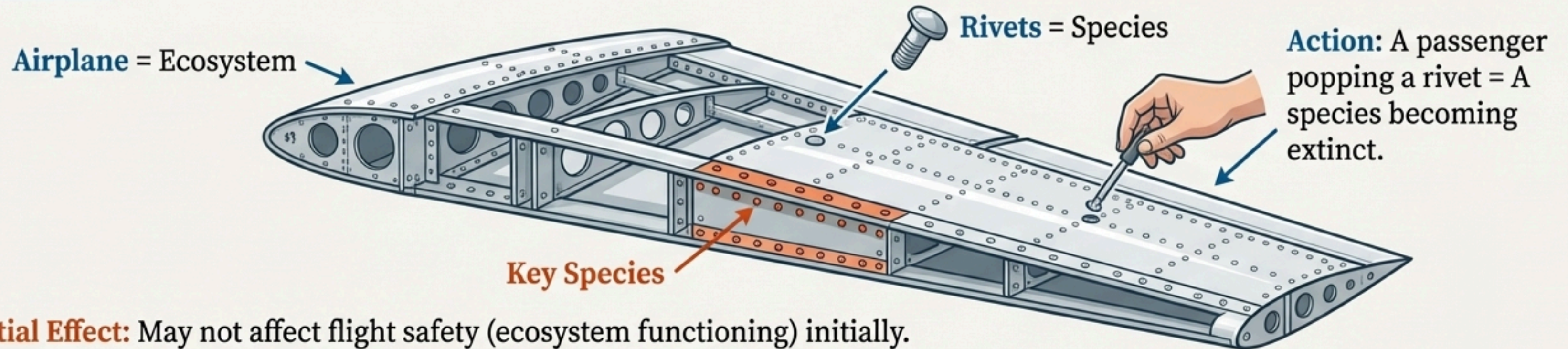
David Tilman's Long-Term Ecosystem Experiments

- **Finding 1:** Plots with more species showed **less year-to-year variation** in total biomass.
- **Finding 2:** Increased diversity contributed to **higher productivity**.

A Stable Community Should:

1. Not show too much variation in productivity.
2. Be resistant or resilient to disturbances.
3. Be resistant to invasions by alien species.

Analogy: The 'Rivet Popper Hypothesis' (Paul Ehrlich)



Initial Effect: May not affect flight safety (ecosystem functioning) initially.

Cumulative Effect: As more rivets are removed, the plane becomes dangerously weak.

Key Species: Loss of rivets on the **wings** (key species driving major ecosystem functions) is a more serious threat than losing rivets on the seats.

13.1.4 Loss of Biodiversity: The Sixth Extinction is Underway

Human Impact Evidence: Colonisation of tropical Pacific Islands by humans led to the extinction of more than **2,000** species of native birds.

IUCN Red List (2004) Data (Last 500 years):

Total Extinctions: **784** species (338 vertebrates, 359 invertebrates, 87 plants).

Notable Recent Extinctions: Dodo (Mauritius), Quagga (Africa), Thylacine (Australia), Steller's Sea Cow (Russia), Three subspecies of tiger (Bali, Javan, Caspian).



Current Threat Levels:

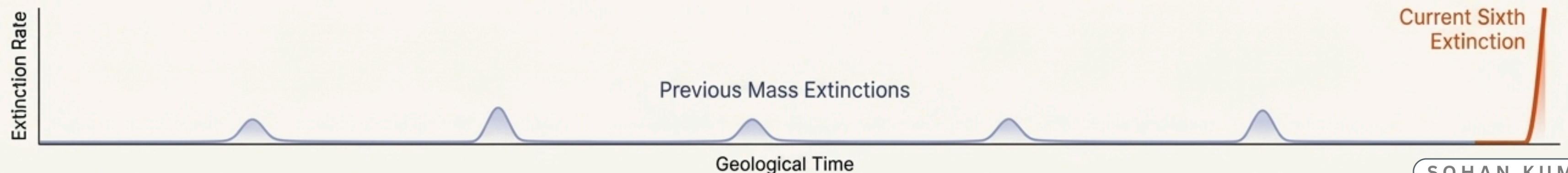
Over **15,500** species worldwide are facing the threat of extinction.

Threatened Groups: 12% of bird species, 23% of mammal species, **32% of all amphibian species**, and 31% of all gymnosperm species.

The Sixth Extinction vs. The Previous Five:

The Difference: The rate of extinction.

Current Rate: Estimated to be 100 to 1,000 times faster than in pre-human times, and human activities are responsible.



The Causes of Biodiversity Loss: 'The Evil Quartet'



1. Habitat loss and fragmentation

The most important cause driving species to extinction.

Tropical rain forests now cover no more than 6% of Earth's land, down from >14%. The Amazon rain forest ('lungs of the planet') is being cleared for cultivating soya beans or for raising beef cattle.



2. Over-exploitation

When human 'need' turns to 'greed'.

Many extinctions in the last 500 years (Steller's sea cow, passenger pigeon) were due to over-exploitation.



3. Alien species invasions

Introduced species become invasive and cause the decline or extinction of indigenous species.

The Nile perch introduced into Lake Victoria led to the extinction of >200 species of cichlid fish.

Invasive weeds like Parthenium (carrot grass), Lantana, and Eichhornia (water hyacinth).



4. Co-extinctions

When a species becomes extinct, plant and animal species associated with it in an obligatory way also become extinct.

When a host fish species becomes extinct, its unique assemblage of parasites also meets the same fate.

13.2.1 Why Should We Conserve Biodiversity?

1. Narrowly Utilitarian: Direct economic benefits.

Products: Food (cereals, pulses, fruits), firewood, fibre, construction material, industrial products (tannins, dyes, resins).

Medicines: Over 25% of drugs sold worldwide are derived from plants. 25,000 species of plants contribute to traditional medicines.

Bioprospecting: Exploring molecular, genetic, and species-level diversity for products of economic importance.

2. Broadly Utilitarian: Ecosystem services.

Oxygen: The Amazon forest produces 20% of the total oxygen in the Earth's atmosphere.

Pollination: Provided by bees, bumblebees, birds, and bats.

Aesthetic Benefits: Pleasures of walking through woods, watching flowers, etc.

3. Ethical Argument: Moral responsibility.

Every species has an intrinsic value, regardless of its economic value to us.

We have a moral duty to care for their well-being and pass on our biological legacy to future generations.

13.2.2 Conservation Methods: In situ (On-site)

Definition Protecting the entire ecosystem to conserve its biodiversity at all levels.
“We save the entire forest to save the tiger.”

Strategy: Biodiversity Hotspots

Regions with very high species richness and a high degree of **endemism** (species confined to that region).

34 biodiversity hotspots in the world. They cover less than **2%** of Earth's land area but could reduce mass extinctions by almost **30%** if strictly protected.

Hotst hotspots covering India:

1. Western Ghats and Sri Lanka
2. Indo-Burma
3. Himalaya



Protected Areas in India

Biosphere Reserves: **14**

National Parks: **90**

Wildlife Sanctuaries: **448**

Sacred Groves

Culturally protected tracts of forest.

Khasi and Jaintia Hills (Meghalaya),
Aravalli Hills (Rajasthan), Western
Ghat regions.

13.2.2 Conservation Methods: Ex situ (Off-site)

Definition: The approach used when a species is endangered or threatened and needs urgent measures to save it from extinction. Threatened organisms are taken out of their natural habitat and placed in special settings.

Traditional Methods



Zoological
Parks



Botanical
Gardens



Wildlife
Safari Parks

Advanced Technologies

- Cryopreservation: Preserving gametes of threatened species in a viable and fertile condition for long periods using low temperatures.
- In vitro fertilization.
- Tissue culture methods for plant propagation.
- Seed Banks: Keeping seeds of different genetic strains of commercially important plants for long periods.

In situ vs. Ex situ Conservation: A Comparative Analysis

Parameter	In situ Conservation	Ex situ Conservation
Meaning	Conservation of species within their natural habitat.	Conservation of species outside their natural habitat.
Approach	Conserves the entire ecosystem.	Conserves a specific organism.
Strategy	Establishes protected areas like National Parks.	Uses special settings like Zoological Parks and Gene Banks.
Best For	Protecting overall biodiversity and ecosystem processes.	Saving critically endangered species from immediate extinction.
Examples (India)	National Parks, Biosphere Reserves, Sacred Groves, Hotspots.	Botanical Gardens, Zoological Parks, Seed Banks, Cryopreservation.

NEET Focus: Assertion-Reason & Statement Analysis

A-R Set 1

Assertion (A): The relationship between species richness and area is a rectangular hyperbola for a wide variety of taxa.

Reason (R): On a logarithmic scale, the species-area relationship is a straight line described by $\log S = \log C + Z \log A$.

Analysis: Both A and R are true, but R is the mathematical representation, not the direct ecological explanation of A.

A-R Set 2

Assertion (A): Introduction of the Nile perch into Lake Victoria led to the extinction of native cichlid fish.

Reason (R): This is an example of habitat loss driving extinction.

Analysis: A is true, but R is **false**. This is an example of an **alien species invasion**.

Statement Analysis

Statement I: The **Western Ghats** are a **biodiversity hotspot** in India.

Statement II: A key criterion for a hotspot is a high degree of **endemism**.

Analysis: Both Statement I and Statement II are true, and Statement II is a reason why regions like the Western Ghats qualify as hotspots.

Rapid Revision: Key Facts & Keywords

Chapter Summary

- Biodiversity exists at 3 levels: Genetic, Species, Ecological.
- Species diversity is highest in the tropics and decreases towards the poles.
- Species-Area relationship slope (Z) is steeper for larger areas (0.6-1.2) than smaller areas (0.1-0.2).
- The Evil Quartet: 1. Habitat Loss, 2. Over-exploitation, 3. Alien Species Invasion, 4. Co-extinction.
- Conservation is of two types: **In situ** (hotspots, national parks) and **Ex situ** (zoos, seed banks).
- Conservation is justified by Narrowly Utilitarian, Broadly Utilitarian, and Ethical arguments.

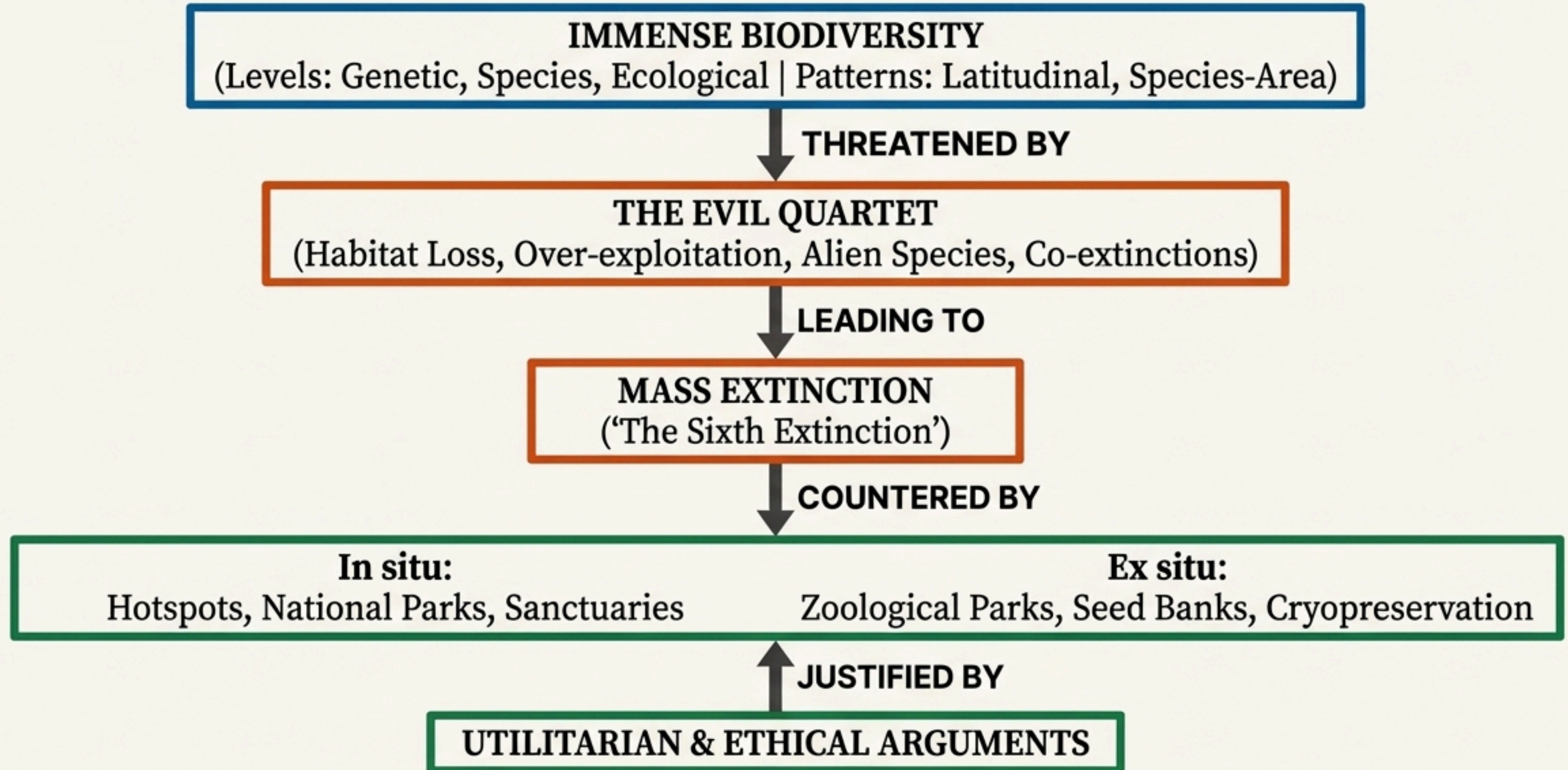
Critical Data

- **Robert May's Estimate:** ~7 million species globally. (Source Serif Pro Regular)
- **IUCN Extinctions (last 500 yrs):** 784 species.
- **Threatened Amphibians:** 32%.
- **Global Hotspots:** 34 total (3 covering India).
- **Amazon Oxygen Production:** 20% of Earth's total.

Keywords List

- **Edward Wilson**
(Source Serif Pro Regular)
- **Rauwolfia vomitoria**
- **Latitudinal Gradient**
- **Species-Area Relationship**
- **Alexander von Humboldt**
- **Z-value**
- **Rivet Popper Hypothesis**
- **The Evil Quartet**
- **Endemism**
- **Biodiversity Hotspots**
- **Sacred Groves**
- **Bioprospecting**
- **Cryopreservation**
- **Earth Summit (Rio, 1992)**
- **World Summit (Johannesburg, 2002)**

Final Chapter Snapshot: The Story of Biodiversity



“If you remember only this much: Earth’s rich **Biodiversity** is threatened by **The Evil Quartet**, forcing a response through **Conservation Efforts** justified by **utilitarian and ethical** reasons.”